

4.3 Contrastive Learning and Zero-Shot Transfer

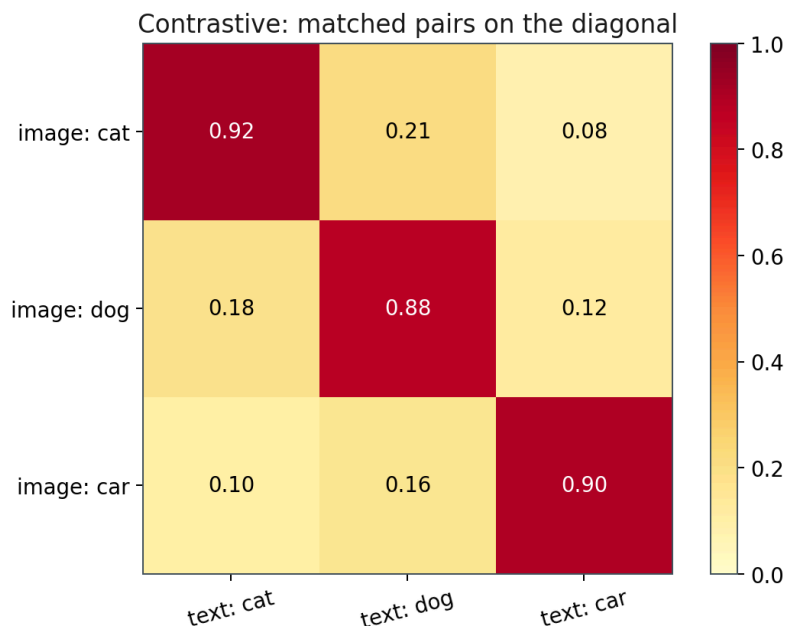
Matched pairs on the diagonal. Class names as text embeddings.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Contrastive learning pulls matched pairs together and pushes unmatched pairs apart, in a shared space. **Zero-shot** transfer is what you get when one side of the pair can be a class name written as text.

1. A batch of matches

N images and N captions. Encode both. Cosine similarities form an $N \times N$ matrix. The loss (InfoNCE) wants the **diagonal** hot: image i with caption i , not caption j .



This is coordinated representation (note **4.1**) with a specific loss. You never fuse pixels and words into one vector during pretraining; you only score pairs.

InfoNCE, one direction, row i :

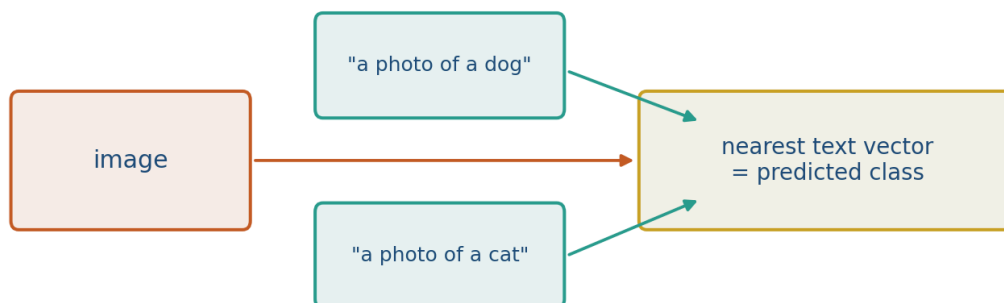
$$-\log \frac{\exp(\text{sim}(v_i, t_i) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(v_i, t_j) / \tau)}.$$

CLIP uses both directions (image-to-text and text-to-image). Temperature τ sharpens the softmax.

2. Zero-shot classification

At test time, encode the image once. Encode prompts "a photo of a dog", "a photo of a cat", ... Take the nearest prompt. There is no trained softmax over ImageNet. New classes are new strings.

Zero-shot: class names are text embeddings, not trained heads



Prompt wording matters (a photo of vs the raw label). So does language: a class that the text tower never saw as a phrase will not magically appear.

3. What can go wrong

The negative set is the rest of the batch. Small batches $\Rightarrow$ easy negatives $\Rightarrow$ a weak space. Web pairs are noisy (Week 5: BLIP filters them). Contrastive spaces can still be **biased** (Week 5.3): "doctor" sits nearer to one gender's photos.

4. Teaching this note

30–40 minutes. Draw the $N \times N$ grid, mark the diagonal, write InfoNCE for $N = 2$, then swap the text side for class prompts. Play the CLIP video on **zero-shot classifiers** and **contrastive training (9:00–22:25)**. Pause on the similarity matrix. Lab 4's tiny InfoNCE is this grid with $N = 4$.

Minute plan: 10 min diagonal vs off-diagonal; 10 min $N = 2$ softmax; 8 min zero-shot prompts; 10 min video. If students ask about negatives in the real paper, the batch was tens of thousands; we will not fake that in lab.

5. Worked example

Two pairs. Cosine matrix (already normalized):

$$S = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}, \quad \tau = 1.$$

Row 0 softmax: $e^{0.9} \approx 2.46$, $e^{0.1} \approx 1.11$, so $[0.69, 0.31]$. InfoNCE term $-\log 0.69 \approx 0.37$.

If every caption is the same vector, every row of S is constant, softmax is uniform, and the diagonal is not special: **the loss cannot learn matching**.

Zero-shot: image $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, prompts dog = $\begin{bmatrix} 0.9 \\ 0.1 \end{bmatrix}$, cat = $\begin{bmatrix} 0.1 \\ 0.9 \end{bmatrix}$. Cosines 0.99 vs 0.10. Predict dog.

6. Where students get stuck

- Treating off-diagonal as “don’t care” instead of **negatives**.
 - Adding a class at test time that the text tower cannot spell (a new visual concept with no words).
 - Using batch size 4 in a real CLIP run and expecting ImageNet-level negatives.
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7. Video

Watch [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Pause on **zero-shot (9:00)** — class names as text — and on **InfoNCE / the $N \times N$ matrix (14:40)**. That matrix is the whole note.

8. Practice

1. If every caption in the batch is the word *photo*, what happens to the diagonal?
2. Why is "a photo of a golden retriever" often better than "golden retriever" for CLIP-style zero-shot?
3. Name one class you **cannot** add at test time by typing it.
4. For $S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\tau = 1$, write the two InfoNCE row-losses (they should be 0). Now change S_{01} to 1. Recompute row 0's softmax and $-\log p_{\text{diag}}$.
5. Three prompts with cosines to one image 0.2, 0.5, 0.1. Which class wins zero-shot? If you divide the cosines by $\tau = 0.07$ before softmax, does the **argmax** change?